THE UNIVERSITY OF AUCKLAND

SEMESTER TWO, 2025
Campus: City

MATHEMATICS

Real Analysis

(Time allowed: TWO hours)

NOTE: Please answer all 8 questions. There are 100 marks available.

The questions are NOT of equal marks.

Show all working and state clearly any theorems that you use in your answers.

Calculators are not permitted in this examination.

Also no books nor loose paper are allowed.

1. (a) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{R}$ and $L \in \mathbb{R}$. Give the definition of $\lim_{n\to\infty} a_n = L$.

(3 marks)

(b) Use the definition of the limit to show that $\lim_{n\to\infty} \dfrac{4n^2}{2n^2 + \cos n} = 2$.

(8 marks)

2. (a) Is the series $\sum(-1)^n \dfrac{n+1}{n^3 + \sqrt{n} + 1}$ convergent?

(6 marks)

(b) What is the radius of convergence of the power series $\sum \dfrac{x^{3n}}{4^n + 5^n}$?

(7 marks)

3. (a) State the intermediate value theorem.

(3 marks)

(b) Let $f\colon [0,\infty) \to \mathbb{R}$ be a continuous function with $f(0) = 3$ and $\lim_{x\to\infty} f(x) = 0$. Prove that there exists a $c \in (0,\infty)$ such that $f(c) = 2$.

(6 marks)

4. (a) Let $A \subset \mathbb{R}$. State what it means for a function $h : A \to \mathbb{R}$ to be Lipschitz continuous.

(3 marks)

(b) Consider the function $h : (0,\infty) \to \mathbb{R}$ given by the formula

$$h(x) := \frac{1}{x}.$$

(i) Prove that h is Lipschitz continuous on $[a,\infty)$, where $a > 0$.

(5 marks)

(ii) Prove that h is not uniformly continuous on $(0,\infty)$.

(6 marks)

5. (a) Use the mean value theorem to prove that

$$|\cos x - \cos y| \leq |x - y|$$

for all $x, y \in \mathbb{R}$.

(5 marks)

(b) Let the function $g : [-1, 1] \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} 0 & \text{for} \quad -1 \leq x < 0 \\ e^x & \text{for} \quad 0 \leq x \leq 1 \ . \end{cases}$$

Explain why there is no function f such that $f'(x) = g(x)$ for all $x \in [-1, 1]$.

(4 marks)

6. (a) Explain why

$$\frac{d}{dx}\left(\int_0^{\sin^2 x} \frac{\log(t+2)}{1+t}\, dt\right)$$

exists, and calculate it. Explain carefully any tools or results that you use.

(6 marks)

(b) Compute

$$\int_{-\pi/4}^{\pi/2} |\sin x| \cos x \, dx$$

mentioning any theorems or results from the notes that you use.

(6 marks)

7. Let $g : [-2, 2] \to \mathbb{R}$ be a bounded function.

(a) Define what is means for P to be a partition of $[-2, 2]$.

(3 marks)

(b) Let P_1 and P_2 be two partitions of $[-2, 2]$.

(i) Explain what this means: Q is a common refinement of the partitions P_1 and P_2.

(3 marks)

(ii) Given arbitrary partitions P_1 and P_2 give an example of a common refinement.

(3 marks)

(iii) Prove that if P_1 and P_2 are two partitions of $[-2, 2]$ then $L(P_2, g) \leq U(P_1, g)$. You may use that if P' is a refinement of any partition P then $U(P', g) \leq U(P, g)$ and $L(P', g) \geq L(P, g)$.

(8 marks)

8. Let $(f_n(x))$ be the sequence of functions on $[0, \infty)$ defined, for each $n \in \mathbb{N}$, by

$$f_n(x) := \frac{x}{x+n}\,.$$

(a) Show that $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0, \infty)$.

(5 marks)

(b) Let $b > 0$. Show that the convergence in part (a) is uniform on $[0, b]$.

(5 marks)

(c) Is the convergence uniform on $[0, \infty)$? Prove your answer.

(5 marks)